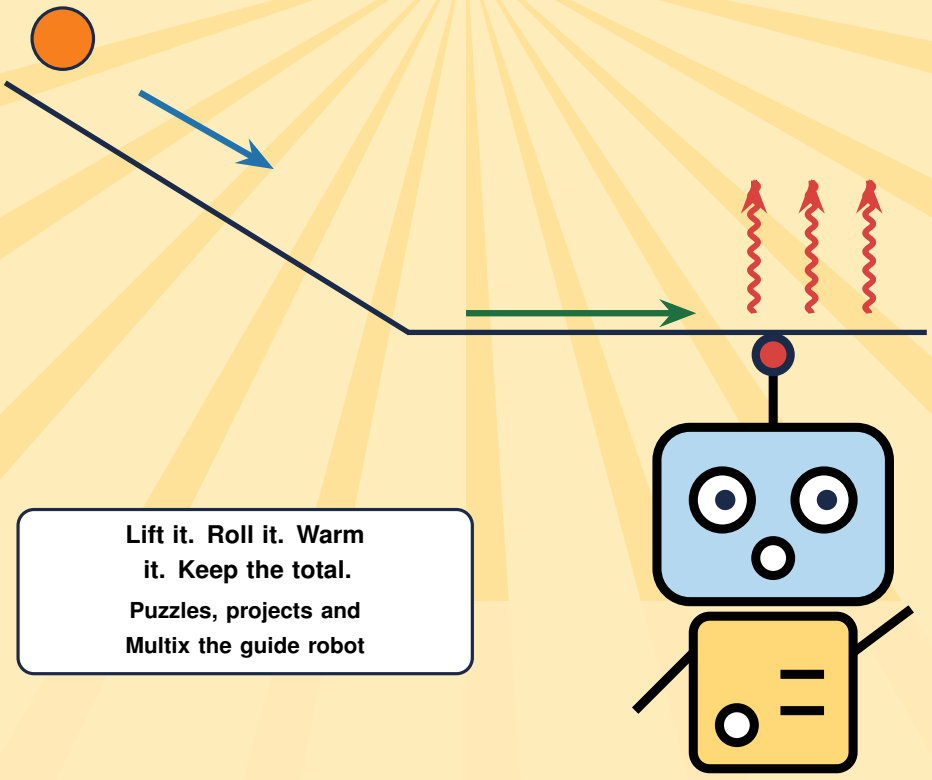
SCIENCE
Book 8

Where the Energy Goes



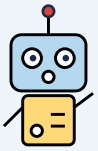
Lift it. Roll it. Warm it. Keep the total.

Puzzles, projects and Multix the guide robot

Ages 9 to 11

This book belongs to _____

Hello. I am Multix.

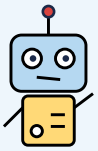


I am a guide robot. I read every instruction very carefully, which is why I sometimes get things wrong. When I make a mistake, see whether you can catch it. I never give away a puzzle answer.

How this book works

Every book in this series asks the same few questions, so once you have read this page you will know your way around.

- **Try it** boxes tell you what to do with a ruler, a cup, some string or a clock. You need nothing from a shop. Everything is at home or at school.
- **What did we keep?** After you simplify something, such as keeping the same conditions in a test or averaging several throws, you say what stayed and what was set aside. Simplifying is fine. Forgetting what you set aside is not.
- **What makes this correction trustworthy?** When you fix or improve an answer, a box says what you checked, what the check gives you reason to trust, and what it does *not* show.
- **Circles** show how a number was made. Page 4 explains how to read them, and Book 1 introduced them.
- **Puzzles** come at the end of each chapter. Try them before you look at the answers on page 45.
- **Slop check** is a page near the end. Slop is hasty, gaudy work that forgot to take care. You will learn to spot it and to fix it.



Here is a fact about me. I watched a ball bounce lower and lower until it stopped. I announced that its energy had been destroyed. Chapter 1 will explain why I should be embarrassed.

In this book

1	Energy You Can Lift	5
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⋆	Words, and notes for grown-ups	47

Try it: Gather your kit

You will use these things across the six chapters. Everything is found at home or at school. Measuring comes from Book 1, fair testing from Book 2, forces from Book 4 and the ledger from Books 5 and 6, and all of them come back here.

- a bouncy ball
- a board and some books
- a ruler and a tape measure
- two cups, one with a lid
- a kitchen thermometer in °C
- a clock or timer
- a rubber band and a bag of coins
- dominoes or small blocks
- a pencil and notebook
- a hard floor and a rug

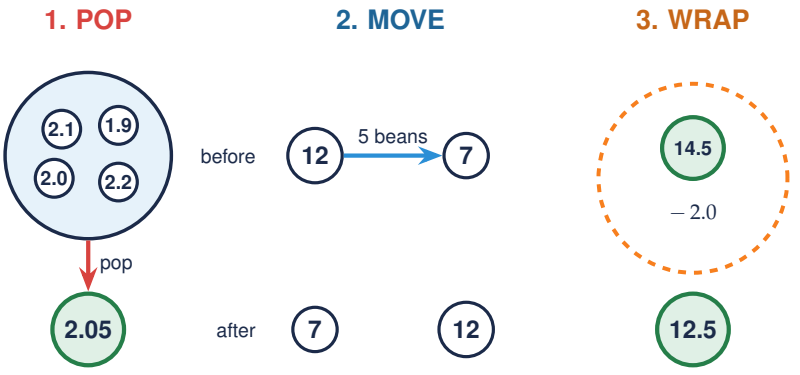
■ **Adult help:** Use warm tap water only, never boiling water from a kettle or stove. Drop things only from low heights and keep ramps low and away from stairs. Keep small balls and coins away from little children and pets. Nothing here needs a flame or mains electricity.

Units in this book

We use the metric system: centimetres (cm), metres (m), seconds (s) and degrees Celsius (°C). Energy is hard to measure at home, so we keep a ledger in made-up **units**: for the same ball, 1 unit is 1 cm of lift. There are 10 mm in 1 cm and 100 cm in 1 m. If you are used to inches, one inch is about 2.5 cm. We will use that only in side notes.

Reading the circles

A circle holds a group of things together. There are three things you can do with circles in this series.

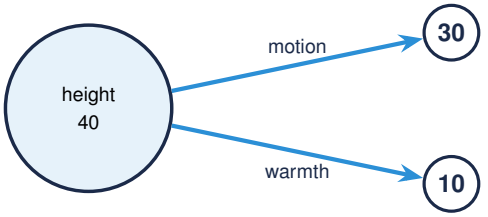


Pop Squash a group down to one value that sums it up, such as an average. A pop is a *summary*, so it always loses something. Every time we pop, we say what the summary keeps and what it loses.

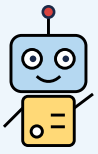
Move An arrow that carries something from one circle to another, such as beans or water. Nothing is summarized. You will see more of moves in later books.

Wrap When you correct something, draw a new dashed circle around the old one and show what changed. The old work stays visible, so anyone can check it.

Energy moves between circles



Energy moves from one circle to others. The total, $30 + 10 = 40$, stays the same.



A pop is not a lie, but it is a shortcut. Any time I tell you a pop, I will tell you what it threw away.

1 Energy You Can Lift

A drop and a bounce, written as heights

Lift a ball and hold it up. It is not moving, yet it can do something the moment you let go: it falls, it speeds up and it hits the floor. We say a lifted ball has **stored energy**, and the higher it is lifted the more it has.

Energy is hard to measure at home. So we use a stand-in. For one ball, the height of the lift tells us how much stored energy it has, and **1 unit is 1 cm of lift**. A ball held at 40 cm has 40 units. This works only for the *same ball*.

Model	What it keeps	What it drops
Height as energy	how much the lift stored	the ball's weight and shape
A bounce height	how much came back up	where the rest went
An energy unit	a number we can add	what energy really is

Try it: Drop and bounce

You need: a bouncy ball, a ruler or tape measure stuck upright to a wall, a hard floor.

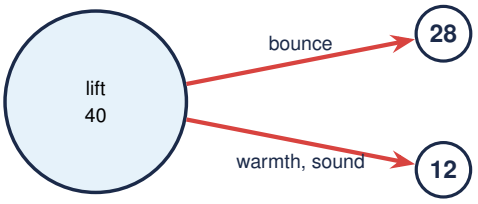
1. Hold the bottom of the ball at 40 cm. Drop it. Do not throw it.
2. Watch how high the bottom of the ball gets on its *first* bounce. Ask a helper to watch too.
3. Repeat five times and write each bounce height. Then work out the average.

	1	2	3	4	5	Average
Bounce (cm)						

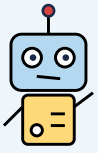
■ **Adult help:** Drop the ball from low heights only and keep clear of windows, stairs and lamps. Small balls can be swallowed by little children and pets.

Practice numbers, not from a real test

Bounces: 29, 27, 28, 28, 28. Total 140, so the average is $140 \div 5 = 28$ cm.
Start: 40 units. Back up: 28 units. The gap is $40 - 28 = 12$ units.
Those 12 units did not vanish. We cannot measure them, but warmth in the ball and floor and the sound of the thud are the likely places.



$28 + 12 = 40$



The ball came back up to 28, not 40, so 12 units were destroyed. I have a very tidy way of saying it. Is it right?

What makes this correction trustworthy?

What we checked. We dropped from the same height five times, with the same ball on the same floor, and averaged the bounces.

What this check gives us reason to trust. Every bounce was within 1 or 2 cm of the others, so the average is steady.

What it does not show. Where the other 12 units went. Warmth and sound are the likeliest places, but our kit cannot measure them.

What did we keep?

We turned a bounce into two heights. **We kept** _____ **and we set aside** _____.

Try it: Three heights

Drop the same ball from 20 cm, 40 cm and 60 cm. Take three bounces at each height and write the average.

Drop (units)	Average bounce (cm)	Lost (units)	Bounce ÷ drop
20			
40			
60			

Practice numbers

Drop 20: bounce 14, lost 6. Drop 40: bounce 28, lost 12. Drop 60: bounce 42, lost 18. $14 \div 20 = 0.7$, $28 \div 40 = 0.7$ and $42 \div 60 = 0.7$. This ball gives back about 7 tenths of its lift. Three tenths go elsewhere.

Puzzle 1.1: Lost

A ball is dropped from 50 and bounces to 35. How many units are lost? _____

Puzzle 1.2: Average

Five bounces were 31, 29, 30, 30 and 30. Average _____ cm. Lost from a 40 drop _____

Puzzle 1.3: Which form?

Write S for stored, M for motion or W for warmth.
(a) ball held up high _____ (b) ball falling _____ (c) floor slightly warm after a hit _____
(d) book on a top shelf _____ (e) rolling skateboard _____

Puzzle 1.4: Unfair test

Ana dropped a tennis ball from 40 and Ben dropped a foam ball from 60. They say foam bounces worse. What is wrong with their test? _____

Puzzle 1.5: Next bounces

A ball bounces to 28, then 20, then 14. How much was lost between the 1st and 2nd bounce? _____ Total lost from a 40 drop to the 3rd bounce? _____

Puzzle 1.6: Fill the row

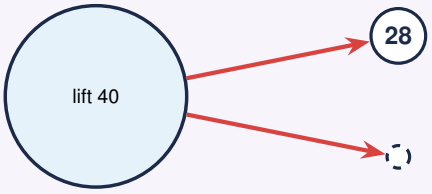
Drop 80, bounce 56. Lost _____ units.

Puzzle 1.7: Which sentence?

(a) “12 units were destroyed.” (b) “12 units went somewhere we did not measure.” Which is better supported? _____

Puzzle 1.8: Draw the circles

A ball starts with 40 and bounces to 28. Fill in the empty circle.



Puzzle 1.9: Wrap

A student wrote “The ball lost energy.” Wrap it with a clearer claim. _____

Puzzle 1.10: Design a drop

Choose a different ball or floor and say what you would keep the same. _____

Bounce record

Try three balls or three floors. Keep the drop height the same each time.

Ball and floor	Drop (units)	Average bounce	Lost	Bounce ÷ drop

Which combination gave back the most? _____

Which lost the most, and what might it have lost it to? _____

Sketch your best bounce and mark the heights.

2

Where a Rolling Ball Goes

Following the energy until the ball stops

A ball at the top of a ramp has stored energy. Let go, and the stored energy turns into **motion**. The ball rolls onto the floor and keeps going, but not forever. It slows and stops. Where did its motion go?

The floor and the ball rub against each other all the way. Rubbing turns motion into **warmth**, a tiny bit at every point. The warmth is too small to feel, so we follow the energy by what we can measure: the height at the start and the distance rolled.

Model	What it keeps	What it drops
A ramp height	the stored energy at the start	the ball's spin
A roll distance	how far the motion lasted	the speed at each point
A surface	how much it rubs	its exact roughness

Try it: Roll it out

You need: a board, books for a ramp 20 cm high, a ball, a tape measure, a bare floor, a rug and a towel.

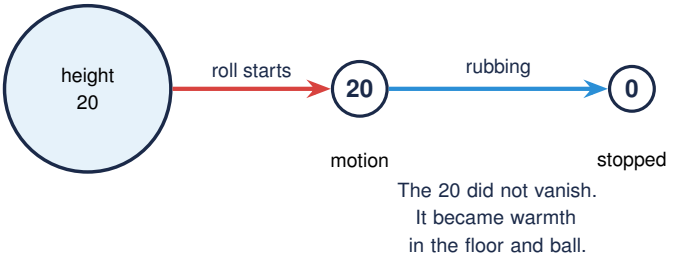
1. Prop the board on books so its top end is 20 cm above the floor.
2. Release the ball from the top without pushing. Measure from the bottom of the ramp to where the ball stops.
3. Do three runs on each surface and average them.

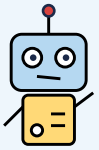
Surface	Run 1	Run 2	Run 3	Average (cm)
Bare floor				
Rug				
Towel				

■ **Adult help:** Keep the ramp low and well away from stairs, and put the books where nobody will trip on them. Wipe any dust from the board first.

Practice numbers, not from a real test

Floor: 148, 152, 150, average 150 cm. Rug: 58, 52, 55, average 55 cm. Towel: 26, 24, 25, average 25 cm.
The same ball started with the same 20 units. On the towel it stopped after 25 cm, on the floor after 150 cm. A rougher surface rubs more, so the motion is used up sooner.





The ball stopped, so its motion is gone. Zero left, nothing to find. I even drew a zero in the circle. Is that all?

What makes this correction trustworthy?

What we checked. We used the same ball and the same ramp height for every surface, and did three runs on each.

What this check gives us reason to trust. The three runs on each surface were close together, so the average is steady. The towel always stopped it sooner than the rug.

What it does not show. How much warmth the ball made, because it is too small to feel. We name it, but we did not measure it.

What did we keep?

We turned a roll into one distance. **We kept** _____ **and we set aside** _____.

Try it: Higher ramp

Use the bare floor. Set the ramp at 10 cm, 20 cm and 30 cm. Take three runs at each height and average them.

Ramp height (units)	Average roll (cm)	Roll ÷ height
10		
20		
30		

Practice numbers

Ramp 10: roll 75 cm. Ramp 20: roll 150 cm. Ramp 30: roll 225 cm.
75 ÷ 10 = 7.5, 150 ÷ 20 = 7.5 and 225 ÷ 30 = 7.5. Each unit of lift gave about 7.5 cm of roll on this floor.

Puzzle 2.1: Average

Three runs on a rug were 62, 58 and 60. Average _____ cm

Puzzle 2.2: Which stops first?

Floor 150, rug 55, towel 25. Which stops the ball soonest? _____ How much shorter than the floor? _____ cm

Puzzle 2.3: Unfair test

Ana rolled a ball from a 10 cm ramp onto the rug. Ben used a 20 cm ramp on the floor. They say the floor is slippery. What is wrong? _____

Puzzle 2.4: Predict

On this floor each unit of lift gives 7.5 cm of roll. Predict the roll from a 40 cm ramp. _____ cm

Puzzle 2.5: Fill the account

A ball starts with 20 units of lift. After it stops, motion is _____ and warmth gained is _____.

Puzzle 2.6: Which circle?

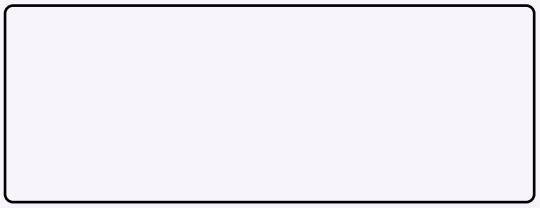
Write S, M or W for stored, motion or warmth.
(a) ball at the top _____ (b) rolling on the floor _____ (c) stopped on the rug _____

Puzzle 2.7: Why sooner?

Why does a ball stop sooner on a rug than on a bare floor?
_____

Puzzle 2.8: Draw the circles

Draw the ball on the floor as circles: height 20, then motion, then warmth.



Puzzle 2.9: Wrap

A student wrote "The rug destroyed the ball's motion." Wrap it with a better claim.
_____

Puzzle 2.10: Design a longer roll

Say two changes that should make a ball roll farther, and one thing you would keep the same. _____

Roll record

Test three balls or three surfaces. Keep the ramp the same each time.

Ball and surface	Run 1	Run 2	Run 3	Average

Which rolled farthest? _____

Which stopped soonest, and what might explain it? _____

Sketch your ramp and mark where each ball stopped.

3 Warm Things Cool Down

Warmth moves from hot to cold

Rub your hands together hard for twenty seconds. They warm up. You did not light a fire. The motion of your hands turned into warmth, so rubbing is a way to change motion into warmth.

Warmth can also move on. A cup of warm water on a table loses warmth to the air and the table until it is as cool as the room. The cup's warmth did not disappear. It went to the room, which got a tiny bit warmer.

Model	What it keeps	What it drops
Rubbing as a change	motion becoming warmth	how each rub heats
A temperature drop	how much warmth left	where in the room it went
A lid	how much warmth it held in	its exact thickness

Try it: Rub and feel

Press your palms together and rub them quickly for 20 seconds. Then hold one palm against your cheek. Now try slow rubbing for 20 seconds and compare.

	Fast rub	Slow rub
Warmth (none, little, lots)		

Try it: Two cups

You need: two identical cups, a thermometer in °C, a clock, warm tap water (never boiling), a lid or plate for one cup.

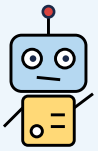
1. Pour the same amount of warm tap water into each cup. Put the lid on one.
2. Read both thermometers at the start and every 5 minutes for 20 minutes.

Minutes	0	5	10	15	20
Open (°C)					
Lidded (°C)					

■ **Adult help:** Use warm tap water only, never water from a kettle or stove. An adult should check the temperature first.

Practice numbers, not from a real test

Open cup: 46, 41, 37, 34, 31. Drops: 5, 4, 3, 3, total 15. Lidded cup: 46, 43, 41, 39, 38. Drops: 3, 2, 2, 1, total 8.
The open cup lost 15 units of warmth and the lidded cup lost 8. The lid slowed the warmth leaving. The difference is $15 - 8 = 7$.



My hands got warm from rubbing, which means I made warmth from nothing. Free heat! I will build a heater. Is that how it works?

What makes this correction trustworthy?

What we checked. We started both cups at the same temperature, used the same amount of water and read them at the same times.

What this check gives us reason to trust. The open cup lost more warmth than the lidded cup in every interval, so the lid made the difference.

What it does not show. Where in the room the warmth went. The room warms by too little for our thermometer to notice.

What did we keep?

We turned cooling into a list of readings. **We kept** _____ **and we set aside** _____.

Puzzle 3.1: Total drop

A cup reads 50, 44, 40, 37 over three intervals. Total drop _____

Puzzle 3.2: Lid or no lid

Both cups start at 48. After 20 minutes the lidded cup reads 40 and the open cup reads 34. How much more did the open cup lose? _____

Puzzle 3.3: From and to

Write the form before and after.
(a) rubbing hands: _____ to _____
(b) a ball starting down a ramp: _____ to _____

Puzzle 3.4: Unfair test

Ana's lidded cup started at 55 and Ben's open cup at 45. She says lids work better. What is wrong? _____

Puzzle 3.5: Where does it go?

A hot cup cools on a table. Name two places its warmth goes. _____

Puzzle 3.6: Cold cup

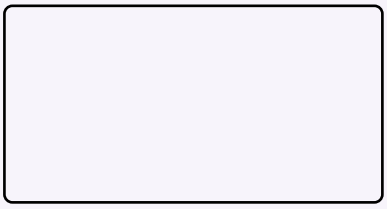
A cold cup warms from 8 to 14. How many units did it gain? _____

Puzzle 3.7: Predict

Which cools faster, a small cup of warm water or a big one? Say why, then say how you would test it. _____

Puzzle 3.8: Draw the circles

A cup drops 15 units. Draw a circle for the cup and a circle for the room.



Puzzle 3.9: Wrap

A student wrote “The cold went into the cup.” Wrap it with a better claim. _____

Puzzle 3.10: Design a keeper

Say how you would keep a drink warm for an hour, and which readings would show it worked. _____

Cooling record

Test three ways of keeping water warm: a lid, a cloth wrapped round, a sock. Use the same start and the same cup.

Cover	Start (°C)	After 20 min	Drop	Rank

Which cover kept the warmth best? _____

How would a repeat make this more trustworthy? _____

Sketch your cups and show where the warmth escapes.

4 Machines Trade Distance

Less push, but over farther

A **lever** is a stiff bar that turns on a point called the pivot. A **pulley** is a wheel with a rope over it. A **ramp** is a slope. These simple machines do not make energy. They let you trade: you push with less force, but you push over a longer distance.

For the lever the trade is easy to see. Multiply the number of coins by their distance from the pivot on each side. When the two products match, the lever balances.

Model	What it keeps	What it drops
Coins × distance	the balance rule	the weight of the ruler
Pull × distance	the trade a pulley makes	friction in the wheel
A machine as a trade	less force for more distance	everything inside it

Try it: Balance a ruler

You need: a 30 cm ruler, a pencil, coins of one kind and sticky tape.

1. Rest the ruler on the pencil at its 15 cm mark (the pivot).
2. Stack 6 coins 2 cm from the pivot on one side. That is $6 \times 2 = 12$.
3. Find where 3 coins balance, then 2 coins, then 1 coin. Write each distance and the product.

Coins on far side	**3**	**2**	**1**
Balance distance (cm)			
Coins × distance			

Practice numbers, not from a real test

Load: 6 coins at 2 cm, so 12. Three coins balanced at 4 cm, giving $3 \times 4 = 12$. Two coins balanced at 6 cm, giving 12. One coin balanced at 12 cm, giving 12.
Fewer coins needed more distance. The product stayed the same. Real balances wobble by a centimetre or so, so real products may differ slightly.

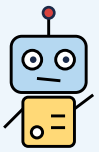
Try it: A pulley

Hang a bag of coins from a string over a pencil or a spool, and lift it 10 cm by pulling straight down. Use a rubber band in the line and write how far it stretches. Then loop the string under a second spool attached to the bag, fix one end above and pull up on the other end. Write the stretch and how far you had to pull the string.

	Stretch (cm)	String pulled (cm)	Stretch × string
Straight		10	
With pulley			

Practice numbers

Straight: stretch 6, string 10, product 60. Pulley: stretch 3.2, string 20, product 64.
The pulley halved the pull but needed twice the string. The product went from 60 to 64. The extra 4 is a gap, and friction in the wheel is a likely place: it turned some motion into warmth.



A lever needs only 2 coins instead of 6. That is less energy overall, so a lever saves energy. I will lever everything. Is that right?

What makes this correction trustworthy?

What we checked. We multiplied each side of the lever and each pull by its distance, and we compared the products.

What this check gives us reason to trust. On the lever the products matched. On the pulley the products were close, and the extra was named as friction.

What it does not show. That a machine saves energy. It saves effort for a moment and costs distance.

What did we keep?

We turned a machine into a product. **We kept** _____ **and we set aside** _____.

Puzzle 4.1: Balance

6 coins sit at 2 cm. Where do 4 coins balance? _____ cm

Puzzle 4.2: Another

8 coins sit at 3 cm. Where do 2 coins balance? _____ cm

Puzzle 4.3: Pulley

Lifting straight takes a pull of 5 over 10 cm, a product of 50. A pulley halves the pull. How far is the string pulled? _____ cm

Puzzle 4.4: Fit the job

Write L for lever, R for ramp, P for pulley or W for wheel.
(a) raise a flag _____ (b) roll a barrel into a truck _____ (c) lift a rock a little way _____ (d) move a cart _____

Puzzle 4.5: The trade

A lever has 8 coins at 5 cm. A helper pushes 20 cm out. How many coins' worth of push is needed? _____ How many times farther does the helper push? _____

Puzzle 4.6: Spot the cheat

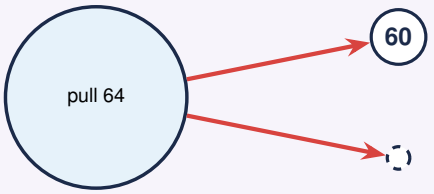
A diagram says a pulley lifts a load 10 cm with half the pull and only 10 cm of string. What is wrong? _____

Puzzle 4.7: Gap

Straight pull product 60. Pulley product 64. Gap _____. Name a likely cause. _____

Puzzle 4.8: Draw the circles

A pulley takes in 64 units. 60 lift the bag. Fill in the circle for the rest.



Puzzle 4.9: Wrap

A student wrote "A lever makes work for free." Wrap it with a better claim. _____

Puzzle 4.10: Design a lifter

Plan a machine that lifts a heavy book with a light push. Say what you would measure. _____

Lever record

Try three loads on the same ruler. Keep the pivot at the middle.

Load	Distance (cm)	Balancing coins	Distance (cm)	Product match?

Which balance was easiest to find? _____

Did any product differ, and by how much? _____

Sketch your lever with its coins and distances.

5 Keeping a Running Total

Adding up what is lost at every step

A bouncing ball loses some of its lift on every bounce. To follow the energy, write the height of each bounce and keep a **running total** of what has been lost so far. The running total grows. The height left plus the total lost always adds back to the start.

Model	What it keeps	What it drops
A bounce list	the height after each bounce	the sound of each thud
A running total	the loss so far	what each loss became
A final check	height left + lost = start	small errors in reading

Try it: Bounce by bounce

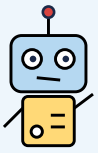
Drop the ball from 40 cm. A helper writes down the top height of bounces 1, 2, 3 and 4 (a phone video helps). Work out the loss at each bounce and the running total.

Bounce	Height	Lost this bounce	Lost so far	Height + lost
Start	40	0	0	40
1				
2				
3				
4				

Practice numbers, not from a real test

Heights: 28, 20, 14, 10.
Lost at each bounce: $40 - 28 = 12$, $28 - 20 = 8$, $20 - 14 = 6$, $14 - 10 = 4$.
Running total lost: 12, then $12 + 8 = 20$, then $20 + 6 = 26$, then $26 + 4 = 30$.
Check: $10 + 30 = 40$. The account balances.

Bounce	Height	Lost now	Lost so far	Check
Start	40	0	0	40
1	28	12	12	40
2	20	8	20	40
3	14	6	26	40
4	10	4	30	40



After bounce 3 I lost only 6 units, so the total lost is 6. Earlier bounces are history and do not count. Right?

What makes this correction trustworthy?

What we checked. We wrote every bounce, and after each one we added the height left to the total lost and checked that it came back to the start.

What this check gives us reason to trust. Every row gave 40, so no unit was dropped from the account.

What it does not show. What each loss turned into. A running total says how much, not into what.

What did we keep?

We turned a bouncing ball into a table. **We kept** _____ **and we set aside** _____.

Try it: On a rug

Repeat on a rug, with the drop at 40 cm.

Bounce	Height	Lost this bounce	Lost so far	Height + lost
1				
2				
3				

Practice numbers

Rug heights: 14, 5, 2. Lost: 26, 9, 3. Running total: 26, 35, 38. Check $2 + 38 = 40$.
The rug took 26 units at the first bounce, against 12 on the floor.

Puzzle 5.1: Running total

Bounces from a 40 drop: 30, 22, 16. Lost each time _____, _____, _____. Total lost _____. Check: 16+ total = _____

Puzzle 5.2: Fill the table

Start 50. Bounces 36, 26, 18. Lost so far after each bounce: _____, _____, _____.

Puzzle 5.3: Biggest loss

Lost this bounce were 12, 8, 6, 4. Which bounce lost the most? _____ Why might it lose most? _____

Puzzle 5.4: Add the losses

Add 12, 8, 6 and 4. _____

Puzzle 5.5: Fix the total

A student wrote the total lost after bounce 3 as 6. What is the right number? _____

Puzzle 5.6: Chain of steps

Start 60. Steps lose 10, 9 and 7. How many are left? _____

Puzzle 5.7: Where did it go?

Name two places a ball's lost energy goes at each bounce. _____

Puzzle 5.8: Draw the chain

Draw circles for a start of 40, a bounce of 28 and a second bounce of 20, with the loss at each step as its own circle.

Puzzle 5.9: Wrap

A student wrote "The ball ran out of energy." Wrap it with a better claim. _____

Puzzle 5.10: Link

A running total is like a savings jar. Say what you add each time and what you check at the end. _____

Bounce ledger

Make a full ledger for a floor of your choice, from a 40 cm drop.

Bounce	Height	Lost now	Lost so far	Height + lost
1				
2				
3				
4				
5				

Does every row come back to 40? _____

Which row has the biggest loss? _____

Draw your running total as a bar chart.

6 Finding Missing Energy

Spotting a gap, and a diagram that cheats

An energy account has a rule: what comes out adds up to what went in. When a number is short, the gap needs a name, such as warmth or sound. When a number is *too big*, something is wrong, because energy cannot appear from nowhere. A good account checks both.

Model	What it keeps	What it drops
Energy as heights	the stored energy of a ball	its weight and shape
A roll or a cooling	what moved or left	where the warmth ended up
A machine as a trade	less force for more distance	what is inside
An energy ledger	where the loss happened	the cause of the loss

Try it: Expected against found

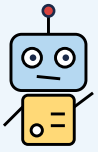
Drop from 40. Use the rule from chapter 1: each bounce gives back about 7 tenths. Expected heights are about 28, 20 and 14. Measure the real bounces and compare. Does any bounce lose more than expected?

Bounce	Expected	Found	Extra lost
1	28		
2	20		
3	14		

Practice numbers, not from a real test

Found: 28, 20, 10. Extra lost: 0, 0 and $14 - 10 = 4$.
The gap appeared at bounce 3. The ball had clipped a chair leg and thumped it, so the extra 4 units went into the chair and into sound.

Bounce	Expected	Found	Extra lost
1	28	28	0
2	20	20	0
3	14	10	**4**



Four units were short at bounce 3, so four units were destroyed. I wrote "destroyed" in the ledger. Neat. Is it?

What makes this correction trustworthy?

What we checked. We compared what the bounce rule expected with what we found at every bounce, so the step with the gap was visible.

What this check gives us reason to trust. Only bounce 3 showed extra loss, and the chair leg matched it.

What it does not show. That every gap has such a visible cause. Gaps of 1 or 2 cm are within our reading error and cannot be traced.

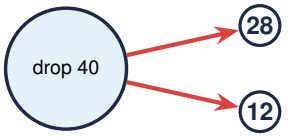
What did we keep?

We turned an energy account into expected and found. **We kept** _____ **and we set aside** _____.

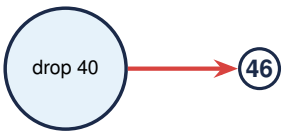
Spot the diagram that cheats

Here are four claims, each drawn as circles. Three balance, or have a named gap. One makes energy from nowhere.

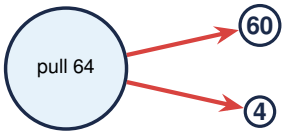
A: ball bounce



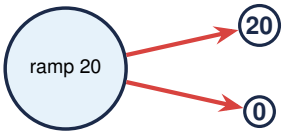
B: ball bounce



C: pulley



D: rolling ball



Which diagram cheats? _____ How many units appeared from nowhere? _____

Which diagram has a gap that is named? _____ What is the name? _____

Puzzle 6.1: Find the step

Start 60. After step 1 expected 42, found 42. After step 2 expected 29, found 24. Which step lost extra? _____ How much? _____

Puzzle 6.2: Too good

A ball dropped from 30 bounced to 36. Does it break the account? _____ Name one thing that might have gone wrong. _____

Puzzle 6.3: Lever check

3 coins at 10 cm are said to balance 2 coins at 10 cm. Do the products match? _____

Puzzle 6.4: Within error?

A bounce was found 1 cm lower than expected, and heights are read to about 1 cm. Can you tell where it went? _____ Why? _____

Puzzle 6.5: Check one step

Why is checking after every bounce better than checking only the last one? _____

Puzzle 6.6: Add the gaps

Extra losses of 0, 2 and 1 at three steps. Total _____

Puzzle 6.7: Model or real?

Write M for model and R for real thing.
(a) a circle with 28 in it _____ (b) the bouncing ball _____ (c) a table of heights _____

Puzzle 6.8: Which question?

The ledger can answer "where did the energy go?" Can it answer "why did the ball clip the chair?" _____ What else would you need? _____

Puzzle 6.9: Unfair ledger

Ana measured bounces on a rug and Ben on a bare floor, and then they added their losses. Why does that mislead? _____

Puzzle 6.10: Recount

Bounces from 40 were 28, 20 and 10. Losses were 12, 8 and _____. Total lost _____

Puzzle 6.11: Wrap

A student wrote "4 units were destroyed." Wrap it with a better claim. _____

Puzzle 6.12: A good habit

Write one habit from this book that makes an energy account more trustworthy. _____

Puzzle 6.13: Missing step

A student measured only the drop and the last bounce, then said all of the loss was "warmth". What could go wrong? _____

Puzzle 6.14: Draw the step circles

Draw a bounce as circles: a drop of 40, a bounce of 28 and the loss.

Puzzle 6.15: Draw the gap

Draw bounce 3 with its 4 unit extra loss as its own circle.

Puzzle 6.16: Draw the wrap

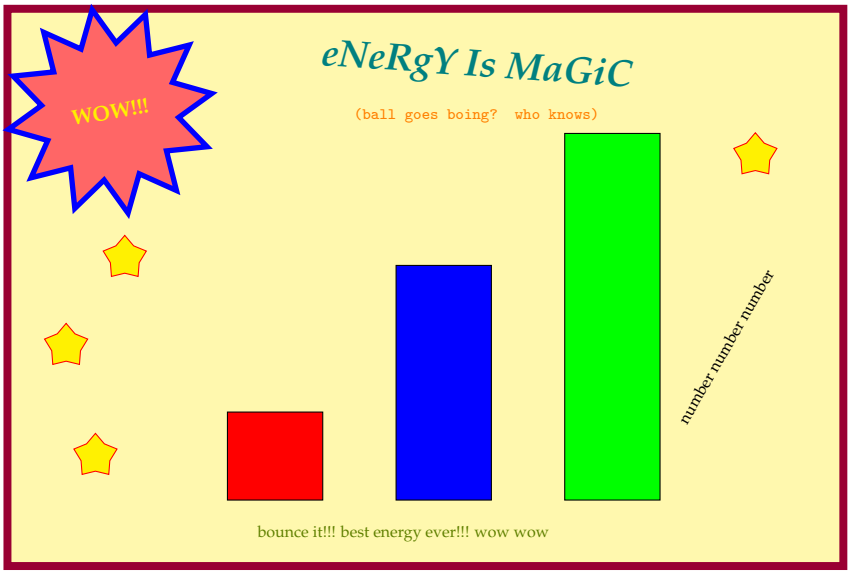
Draw a circle around “4 units destroyed” and write the corrected claim outside it.

Slop check

Slop is hasty, gaudy work that forgot to take care. It can be made by a person or a machine, quickly or slowly. What matters is whether the work shows care for three things:

- **Salient:** Is the main point easy to see?
- **Simple:** Is there anything that does nothing but get in the way?
- **Relevant:** Does everything on the page help answer the question?

Here is a hasty poster about a bouncing ball. Study it, then decide what is wrong.



Slop check

(a) What is the poster's main point? Can you tell?

_____

(b) Name two things that are not simple, because they only distract.

_____

(c) Name two things that are missing, so the bars cannot be read.

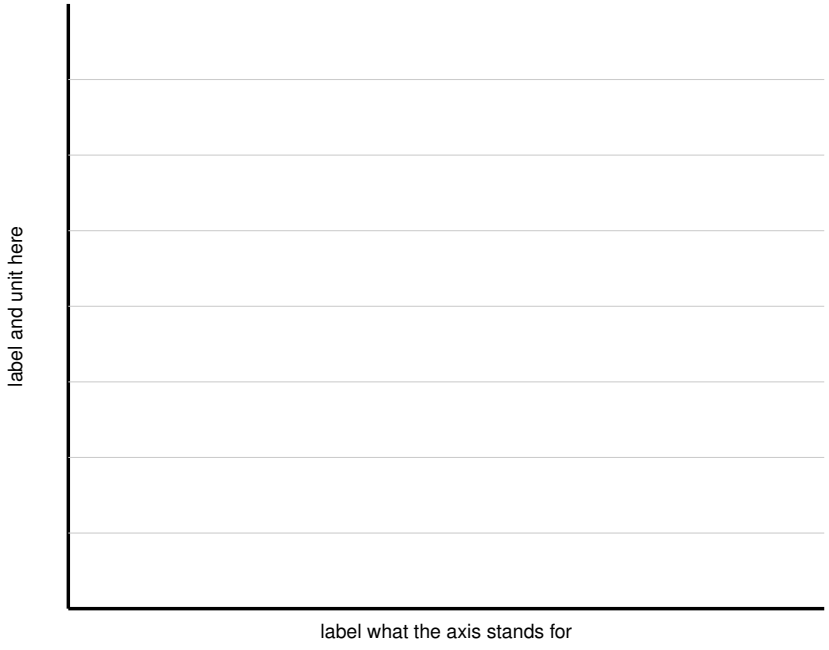
_____

(d) Which part is not relevant to a bouncing ball at all?

_____

Now you try

On a separate sheet, redraw a clear bar chart of the ball's bounce heights: drop 40, bounce 1 at 28, bounce 2 at 20 and bounce 3 at 14, all in cm. Give it a title, label the axes with units, use one or two calm colours, and delete everything that does not help. Check it against the three words: salient, simple, relevant.

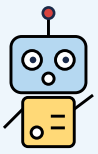


Title:

What did we keep, across the whole book?

For each chapter, say one thing that was simplified, what was kept, and what was set aside. A sample is done for you.

Chapter	What we simplified	What we kept	What we set aside
1 Energy you can lift	a bounce as two heights	the drop and the bounce	sound and warmth, not measured
2 A rolling ball	a roll as one distance	the roll for a ramp height	the rubbing at each point
3 Warm things cool	cooling as readings	start, end and the drop	where in the room it went
4 Machines	a machine as a product	effort times distance	friction inside
5 Running total	bounces as a list	the total lost so far	what each loss became
6 Missing energy			



Every number in this book stood in for energy that we could not measure directly. A stand-in is allowed to drop things, as long as you say which. Next time something seems to stop, you will know to ask where it went.

Write an energy report

1. **What I looked at.**
2. **What I changed, measured and kept the same.**
3. **My readings.** All of them.
4. **My answer.** With the gap named.
5. **What I kept and what it hides.**

Pick one test from this book and write a report on it. Use the lines below.

Tests to try

Pick one and design it with the habits from this book: change one thing, keep the rest the same, repeat, and keep a running total.

1. Does a ball bounce higher from a table or from the floor?
2. Does a heavier ball roll farther down the same ramp?
3. Which cover keeps a warm cup warm the longest?
4. Where on a ruler does one coin balance two coins?
5. How many bounces does a ball make before it is under 5 cm?
6. Does a longer ramp, at the same height, change how far a ball rolls?

Write the test you pick:

_____

_____

_____

_____

_____

_____

Change: _____ Measure: _____ Keep the same: _____

A week of energy

Each day, notice one place where energy changes form and say from what to what.

Day	What I saw	From and to
Mon		
Tue		
Wed		
Thu		
Fri		
Sat		
Sun		

Questions I still have

Write what you still wonder about energy. Then pick one and think how to test it.

Answers

Try every puzzle first. Where an answer shows a calculation, check the steps, not just the result.

Chapter 1

Multix on the 12 units. No. They went somewhere we did not measure, most likely warmth and sound. "Destroyed" claims more than we know. **Keep box.** Kept: the drop height and the bounce height. Set aside: sound and warmth, not measured.

1.1 $50 - 35 = 15$.
1.2 $150 \div 5 = 30$ cm. Lost $40 - 30 = 10$.
1.3 (a) S (b) M (c) W (d) S (e) M.
1.4 Different balls and different drop heights, so the ball could not be the only reason.
1.5 $28 - 20 = 8$. $40 - 14 = 26$.
1.6 $80 - 56 = 24$.
1.7 (b).
1.8 The empty circle is 12.
1.9 "About 12 of the ball's 40 units went into sound and warmth."
1.10 Answers vary. Keep the drop height and the floor the same.

Chapter 2

Multix on the zero. No. The motion became warmth in the floor and ball, too small to feel. The zero shows the motion is gone, not the energy. **Keep box.** Kept: the roll distance for a ramp height. Set aside: the rubbing at each point.

2.1 $180 \div 3 = 60$ cm.
2.2 Towel. $150 - 25 = 125$ cm.
2.3 The ramp height and the surface both differed, so the floor and the ramp could not be separated.
2.4 $40 \times 7.5 = 300$ cm.
2.5 Motion 0, warmth gained 20 (about, in the ledger's units).
2.6 (a) S (b) M (c) W.
2.7 A rug rubs more, so the motion becomes warmth sooner.
2.8 Circles: 20, then 20 as motion, then 0 motion and 20 warmth.
2.9 "On the rug, the ball's motion turned into warmth sooner than on the floor."
2.10 Answers vary. A higher ramp or a smoother surface roll farther. Keep the ball the same.

Chapter 3

Multix on free heat. No. The warmth came from the motion of the hands, and that came from your body's own energy. Nothing was made from nothing. **Keep box.** Kept: start and end readings. Set aside: where in the room the warmth went.

3.1 $50 - 37 = 13$.
3.2 Open lost 14, lidded lost 8. $14 - 8 = 6$ more.
3.3 (a) motion to warmth (b) stored to motion.
3.4 The starts differed by 10, so a lid was not the only difference.
3.5 The air and the table.
3.6 $14 - 8 = 6$.
3.7 Usually the small cup, since there is less water to hold the warmth. Test both with the same start and lid.
3.8 Cup circle 15, room circle gains the same 15.
3.9 "Warmth left the cup and went into the surroundings."
3.10 Answers vary. A lid and a wrap, with readings every 10 minutes.

Chapter 4

Multix on saving energy. No. A lever needs less push but the push is over a longer distance, so the product stays the same or grows. **Keep box.** Kept: effort times distance. Set aside: friction inside.

4.1 $12 \div 4 = 3$ cm.
4.2 $8 \times 3 = 24$. $24 \div 2 = 12$ cm.
4.3 Half the pull is 2.5. $50 \div 2.5 = 20$ cm.
4.4 (a) P (b) R (c) L (d) W.
4.5 $8 \times 5 = 40$. $40 \div 20 = 2$ coins. The helper pushes 4 times farther.
4.6 Half the pull needs twice the string, 20 cm.
4.7 Gap 4. Friction in the wheel.
4.8 The empty circle is 4.
4.9 "A lever trades a smaller push for a longer push."
4.10 Answers vary. Measure the pull and the distance, and compare the products.

Chapter 5

Multix on the running total. No. The total lost is $12 + 8 + 6 = 26$. Earlier losses count. **Keep box.** Kept: the total lost so far. Set aside: what each loss became.

5.1 10, 8, 6. Total 24. $16 + 24 = 40$.
5.2 14, 24, 32.
5.3 Bounce 1. The ball has the most energy there, so there is more to lose.
5.4 $12 + 8 + 6 + 4 = 30$.
5.5 26.
5.6 $60 - 26 = 34$.
5.7 Warmth and sound.
5.8 Circles: 40; 28 with 12; then 20 with 8.
5.9 "After 4 bounces the ball had 10 units left and 30 had gone into warmth and sound."
5.10 Answers vary. You add each deposit and check the jar's total.

Chapter 6

Multix on "destroyed". No. The ledger placed the gap at bounce 3 and a chair leg matched it. The energy went into the chair and into sound. **Keep box.** Kept: where the loss happened. Set aside: the cause of the loss.

Spot the cheat. B. 6 units appeared from nowhere. C has a named gap, friction (warmth).
6.1 Step 2. $29 - 24 = 5$.
6.2 Yes. Energy appeared from nowhere. The ball may have been pushed down, or the heights misread.
6.3 No. $3 \times 10 = 30$ and $2 \times 10 = 20$.
6.4 No. A 1 cm gap is within reading error.
6.5 Checking each bounce shows the step where a gap appeared.
6.6 $0 + 2 + 1 = 3$.
6.7 (a) M (b) R (c) M.
6.8 No. The ledger shows where, not why. You would need to watch or repeat the drop.
6.9 The floors differ, so losses are not comparable.
6.10 $20 - 10 = 10$. $12 + 8 + 10 = 30$.
6.11 "4 extra units were lost at bounce 3 and went into the chair and sound."
6.12 Check after every step, use the same ball and floor, or name each gap.
6.13 Calling the whole loss "warmth" hides a bounce that lost extra to something else.
6.14 to 6.16 Answers vary. Check that every circle's number is accounted for.
Slop check. (a) You cannot tell: no title, no labels. (b) Many colours, stars, joke text. (c) A title, the heights and the units. (d) The stars and jokes.

Words

energy what lets something move, warm up or lift; it changes form but the total stays.

friction rubbing between two things that turns motion into warmth.

ledger a list of numbers that adds up to a total.

lever a stiff bar that turns on a point called the pivot.

model a simple stand-in that keeps some things and drops others.

pulley a wheel with a rope that changes the direction or size of a pull.

ramp a slope that trades distance for effort.

stored energy energy a thing has because of where it is, like a lifted ball.

Notes for grown-ups

This book teaches that an energy account is trustworthy when what came out adds up to what went in and any gap is named. Each chapter has one activity, one worked example with made-up practice numbers, one trust box, one "what did we keep?" prompt and a page of puzzles.

Materials and safety. A bouncy ball, a board and books, a ruler and tape measure, two cups and a lid, a thermometer, coins, a rubber band, string and spools. Use warm tap water only. Keep ramps low and away from stairs. Keep small balls and coins away from young children and pets.

Energy units. Energy is hard to measure at home. The book uses the lift height of the same ball as a stand-in, with 1 unit for each centimetre. The stand-in works only for one ball and one test. Gaps are named, not measured.

The practice numbers were made up so the arithmetic is easy to check. Real results will vary with the ball, the floor and the equipment.

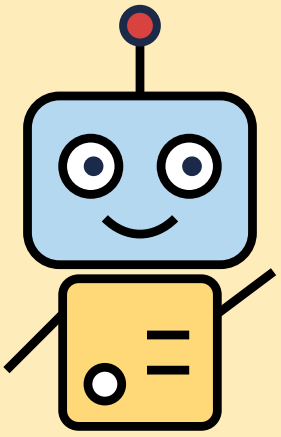
Sources

- R. P. Feynman, R. B. Leighton and M. Sands. *The Feynman Lectures on Physics*, Volume I, Chapter 4, "Conservation of Energy". For the idea of an energy account.
- P. G. Hewitt. *Conceptual Physics*. Pearson. For simple machines and for energy changing form.
- J. R. Taylor. *An Introduction to Error Analysis*, 2nd edition. University Science Books, 1997. For measurement error and repeats.

Next in Science

Small Models, Big Things

A map is smaller than a street. A model is smaller than a room. Book 9 builds one, lists what it keeps and what it drops, and repairs it when it gives a wrong answer.



Ages 9 to 11 ⋆ Science Book 8